Paleoceanography and Paleoclimatology

Supporting Information for

Stable isotope constraints on marine productivity across the Cretaceous–Paleogene mass extinction

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Introduction

This supporting information file contains the geochemical and stable isotope data used in this publication. This includes sample name, depth with respect to the Cretaceous-Paleogene (K/Pg) boundary (cm), percentage of calcium carbonate (%CaCO3), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$), stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$), stable nitrogen isotope composition of bulk organic matter ($\delta^{15}N$), and stable carbon isotope composition of phytane ($\delta^{13}C_{phytane}$). The standard deviation for each parameter is reported when multiple analyses of the same sample were performed. Analytical reproducibility (1σ) was calculated as better than 0.1% for TOC and TN, 0.1‰ and 0.5‰ for $\delta^{13}C_{org}$ and $\delta^{15}N$, and 0.04‰ and 0.09‰ for $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ by comparison with analytical standards (see main text for details).

Sample	cm	%$CaCO_3$	%TOC	st. dev.	$\delta^{13}C_{carb}$	$\delta^{18}O_{carb}$	$\delta^{13}C_{org}$	st. dev.	$\delta^{15}N$	st. dev.	$\delta^{13}C_{phytane}$	st. dev.
KEF11.15	-20	42.54	0.25	0.004	0.94	-3.77	-25.30	0.202	3.20	0.655	-30.70	0.310
KEF11.30	-5	39.37	0.25	0.004	0.87	-3.64	-25.33	0.213	3.58	0.779	-30.67	0.303
KEF12.00	-1	1.82	0.61	0.011	1.44	-3.71	-26.92	0.106	4.53	0.840	-31.95	0.524
KEF12.05	5	3.60	0.76	0.002	0.60	-3.84	-27.19	0.080	4.70	0.539	-33.88	0.392
KEF12.10	10	3.91	0.65	0.044	-0.88	-4.27	-27.20	0.140	4.78	0.100	-31.61	0.430
KEF12.20	20	5.47	0.48	0.004	-1.22	-4.42	-26.32	0.055	5.77	0.151	-31.38	0.799
KEF12.70	70	10.63	0.54	0.027	-1.38	-3.96	-26.35	0.186	4.29	0.316	-32.13	0.035
KEF13.25	125	10.89	0.82	0.015	-1.76	-3.27	-26.49	0.046	5.77	0.158	-31.40	0.157
KEF13.75	175	16.61	1.07	0.059	-1.75	-2.48	-26.08	0.060	5.32	0.285	-31.99	0.246

Table S1. El Kef, Tunisia. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰), stable nitrogen isotope composition of bulk organic matter ($\delta^{15}N$; ‰), and stable carbon isotope composition of phytane ($\delta^{13}C_{phytane}$; ‰).

Sample	cm	%$CaCO_3$	%TOC	st. dev.	$\delta^{13}C_{carb}$	$\delta^{18}O_{carb}$	$\delta^{13}C_{org}$	st. dev.	$\delta^{15}N$	st. dev.
STWA-5-7	-6	38.01	0.77	0.014	0.79	-4.34	-26.20	0.298	5.86	0.583
STWA-0-3	-1.5	33.14	0.87	0.007	1.10	-4.23	-26.40	0.214	6.86	0.155
STWA+1	1	1.33	0.86	0.017	1.13	-3.76	-26.61	0.167	5.92	0.227
STWA+3	2	2.45	0.84	0.044	0.72	-3.38	-26.92	0.163	6.43	0.283
STWA+5	3	2.73	0.93	0.025	-0.03	-3.20	-26.92	0.070	6.68	0.849
STWA+3+5	4	7.00	0.81	0.003	-0.76	-3.65	-26.67	0.239	6.58	0.184
STWA+8+10	9	5.04	0.65	0.017	-1.97	-3.19	-26.24	0.149	7.23	0.551
STWA+20+22	21	11.30	0.72	0.008	-2.52	-2.56	-26.14	0.029	7.41	0.333
STWA+30+32	31	10.43	1.04	0.039	-0.98	-3.72	-26.12	0.190	7.76	0.218
STWA+84	84	31.09	1.58	0.032	-1.17	-2.86	-26.56	0.181	7.76	0.000

Table S2. Aïn Settara, Tunisia. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰), and stable nitrogen isotope composition of bulk organic matter ($\delta^{15}N$; ‰).

Sample	cm	%$CaCO_3$	%TOC	st. dev.	$\delta^{13}C_{carb}$	$\delta^{18}O_{carb}$	$\delta^{13}C_{org}$	st. dev.
Caravaca -20-25	-22.50	80.33	0.15		1.69	-2.44	-23.63	
Caravaca-10-12	-11.00	79.40	0.31	0.015	2.01	-2.37	-23.47	0.049
Caravaca -5-10	-7.50	79.25	0.15		1.64	-2.66	-22.80	
Caravaca-1-3	-2.00	77.40	0.36	0.013	1.98	-2.44	-23.66	0.154
Caravaca+0.5+2	1.25	41.17	0.45	0.024	0.24	-3.63	-25.37	0.279
Caravaca+2+3	2.50	41.93	0.33	0.009	0.05	-3.24	-24.67	0.171
Caravaca+3+4	3.50	42.85	0.31	0.003	0.34	-3.15	-24.40	0.351
Caravaca+4+5	4.50	43.94	0.33	0.010	0.55	-3.05	-24.44	0.113
Caravaca+5+6	5.50	53.64	0.35	0.009	1.02	-2.13	-24.12	0.200
Caravaca+6+7	6.50	69.47	0.37	0.005	1.17	-1.83	-23.87	0.094
Caravaca+7+8	7.50	68.23	0.41	0.041	1.14	-1.82	-23.51	0.045
Caravaca+8+9	8.50	67.00	0.46	0.076	1.13	-1.83	-23.66	0.231
Caravaca+9+10	9.50	75.30	0.36	0.006	1.02	-2.01	-23.80	0.118
Caravaca+11+13	12.00	75.49	0.38	0.006	1.08	-2.08	-24.06	0.004
Caravaca+14+16	15.00	83.79	0.38	0.009	1.20	-1.70	-23.67	0.044
Caravaca+16+18	17.00	83.02	0.35	0.009	1.19	-1.91	-22.83	0.240
Caravaca+20	19.00	87.43	0.39	0.008	1.29	-1.98	-22.87	0.103

Table S3. Caravaca, Spain. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), and stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰).

Sample	cm	%$CaCO_3$	%TOC	st. dev.	$\delta^{13}C_{carb}$	$\delta^{18}O_{carb}$	$\delta^{13}C_{org}$	st. dev.	$\delta^{13}C_{phytane}$	st. dev.
AGOST-22	-22	80.20	0.24	0.013	1.86	-3.02	-24.26	0.100	-31.25	0.753
AGOST-5	-17.5	81.87	0.09		1.50	-2.56			-31.20	0.741
AGOSTT+0+1	0.5	80.20	0.24	0.013	1.86	-3.02	-24.26	0.100	-31.72	0.360
AGOSTT+1+2	1.5	31.55	0.40	0.010	-0.16	-4.06	-22.92	0.078	-30.25	0.741
AGOSTT+2+3	2.5	35.30	0.32	0.018	-0.19	-3.80	-23.08	0.093	-30.64	0.690
AGOSTT+3+4	3.5	33.52	0.34	0.021	-0.18	-3.77	-22.57	0.099	-30.45	0.334
AGOSTT+4+6	5	33.59	0.34	0.020	-0.27	-3.14	-23.19	0.012	-31.48	0.226
AGOSTT+8+10	9	69.54	0.29	0.004	0.81	-2.30	-23.90	0.098	-30.36	0.305
AGOSTT+10+12	11	59.01	0.22	0.000	0.40	-2.21	-23.06	0.086	-30.56	1.088
AGOSTT+12+14	13	81.75	0.41	0.000	0.79	-2.32	-23.97	0.080	-30.94	0.175
AGOSTT+14+16	15	71.21	0.31	0.024	0.68	-2.47	-23.91	0.031	-30.33	21.818
AGOSTT+16+18	17	84.47	0.48	0.004	1.23	-2.10	-23.96	0.082	-30.52	0.371
AGOSTT+18+20	19	78.00	0.28	0.019	0.91	-2.39	-23.86	0.062	-30.93	0.089

Table S4. Agost, Spain. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰), and stable carbon isotope composition of phytane ($\delta^{13}C_{phytane}$; ‰).

Sample	cm	%$CaCO_3$	%TOC	st. dev.	$\delta^{13}C_{carb}$	$\delta^{18}O_{carb}$	$\delta^{13}C_{org}$	st. dev.
Bidart-25	-25	66.04	0.08		1.50	-1.62		
Bidart-6-10	-10	38.99	0.25	0.013	1.86	-1.24	-23.94	0.145
Bidart-0-2	-2	64.47	0.26	0.014	1.47	-2.09	-24.10	0.092
Bidart+0+2	1	35.79	0.27	0.002	-1.57	-3.28	-23.92	0.043
Bidart+3+6	4.5	37.08	0.28	0.010	-1.36	-2.98	-23.97	0.021
Bidart+6+9	7.5	55.70	0.34	0.019	0.46	-1.25	-23.93	0.009
Bidart+9+13	11	51.96	0.29	0.001	0.48	-1.14	-24.02	0.049
Bidart+13+17	15	51.83	0.18	0.000	0.42	-1.45	-24.06	0.024
Bidart+17+21	19	49.96	0.29	0.016	0.35	-1.30	-23.84	0.104
Bidart+21+25	23	65.99	0.29	0.001	0.93	-0.95	-24.06	0.183
Bidart+25+29	27	66.54	0.29	0.015	1.09	-0.84	-23.97	0.088
Bidart+50	50	85.63	0.28	0.022	1.80	-0.39	-24.25	0.139

Table S5. Bidart, France. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), and stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰).

Sample	cm	%$CaCO_3$	%TOC	st. dev.	$\delta^{13}C_{carb}$	$\delta^{18}O_{carb}$	$\delta^{13}C_{org}$	st. dev.
LOYA-25	-25	65.55	0.06		1.13	-2.63		
LOYA-6-10	-8	53.64	0.23	0.010	1.36	-3.12	-24.08	0.150
LOYA-2-6	-4	66.00	0.24	0.020	1.51	-2.92	-24.23	0.090
LOYA-0-2	-1	65.71	0.22	0.000	1.46	-3.59	-24.06	0.130
LOYA+0+3	1.5	65.85	0.24	0.010	1.30	-5.41	-24.61	0.080
LOYA+3+5	4	29.64			0.58	-2.72	-24.69	0.030
LOYA+5+8	6.5	51.88	0.26	0.010	0.67	-2.59	-24.31	0.060
LOYA+8+11	9.5	54.70	0.26	0.020	0.35	-2.75	-24.42	0.020
LOYA+11+14	12.5	50.30	0.26	0.010	0.80	-2.62	-24.30	0.090
LOYA+14+18	16	53.54	0.27	0.020	0.65	-2.82	-24.31	0.040
Loya+18+22	20	52.98	0.27	0.010	1.27	-2.51	-24.25	0.030
Loya+22+26	24	64.19	0.27	0.014	1.27	-2.51	-24.25	0.032

Table S6. Loya, France. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), and stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰).

Sample	cm	%$CaCO_3$	%TOC	$\delta^{13}C_{carb}$	st. dev.	$\delta^{18}O_{carb}$	st. dev.	$\delta^{13}C_{org}$	st. dev.	$\delta^{15}N$	st. dev.
KU -1	-5	96.34	0.10	1.33	0.036	-1.41	0.090	-26.41	0.906	4.91	0.437
KU 0	-0.5	82.36	0.13	1.35	0.000	-1.14	0.000	-25.26	0.409	6.96	2.667
KU 1	0.5	12.67	0.37	1.33	0.068	-1.61	0.224	-25.28	0.247	0.41	0.844
KU b	1.1	20.76	1.55	1.80	0.000	-1.06	0.000	-26.15	0.726	-2.28	1.644
KU 2.1	1.45	52.17	0.95	1.88	0.000	-1.30	0.000	-27.02	0.198	-0.30	0.206
KU 2.2	1.95	55.63	0.92	1.81	0.000	-1.15	0.000				
KU 2.3	2.45	58.33	0.77	1.74	0.000	-1.39	0.000				
KU 2.4	2.95	63.13	0.63	1.75	0.025	-1.22	0.067	-25.70	0.767	-0.51	0.149
KU 2.5	3.45	61.01	1.12	1.63	0.000	-1.37	0.000				
KU 2.6	3.95	63.30	1.18	1.72	0.000	-1.09	0.000	-25.69	0.509	0.49	0.215
KU 2.7	4.45	62.90	1.32	1.71	0.000	-1.22	0.000				
KU 2.8	4.95	64.03	1.04	1.70	0.035	-1.16	0.131	-25.61	0.933	1.68	0.230
KU 2.9	5.3	64.71	1.01	1.70	0.000	-1.03	0.000				
KU 3.1	5.65	60.23	0.67	1.62	0.013	-1.12	0.100				
KU 3.2	6.15	65.47	0.61	1.80	0.000	-0.87	0.000				
KU 3.3	6.65	65.45	0.62	1.57	0.000	-1.27	0.000	-24.58	0.040	5.16	0.581
KU 3.4	7.15	66.58	0.56	1.71	0.000	-1.11	0.000				
KU 3.5	7.65	66.72	0.63	1.69	0.000	-1.09	0.000				
KU 3.6	8.15	68.48	0.66	1.62	0.000	-1.19	0.000				
KU 3.7	8.65	68.34	0.65	1.66	0.044	-1.16	0.037	-24.45	0.193	6.94	2.018
KU 3.8	9.15	71.37	0.61	1.62	0.000	-1.08	0.000				
KU 3.9	9.65	72.36	0.56	1.64	0.000	-1.07	0.000				
KU 3.10	10.15	72.05	0.55	1.59	0.000	-0.96	0.000	-25.09	0.738	5.34	0.736
KU 3.11	10.65	73.16	0.49	1.54	0.000	-1.17	0.000				
KU 4.1	11.15	69.54	0.70	1.65	0.000	-1.12	0.000				
KU 4.2	11.65	72.26	0.66	1.57	0.000	-1.23	0.000				
KU 4.3	12.15	69.43	0.99	1.68	0.000	-1.01	0.000				
KU 4.4	12.65	72.30	0.63	1.57	0.000	-1.21	0.000				
KU 4.5	13.15	71.39	0.58	1.65	0.014	-1.00	0.039	-26.22	0.631	6.02	1.612
KU 4.6	13.65	73.27	0.45	1.61	0.000	-0.96	0.000				
KU 4.7	14.15	73.70	0.43	1.54	0.000	-1.14	0.000				
KU 4.8	14.65	74.80	0.41	1.62	0.000	-0.99	0.000				
KU 4.9	15.15	74.21	0.36	1.53	0.000	-1.21	0.000				
KU 4.10	15.65	73.52	0.51	1.66	0.000	-0.98	0.000				
KU 4.11	16.15	74.71	0.34	1.64	0.041	-1.08	0.085	-25.96	0.099	7.56	1.684
KU 4.12	16.65	74.98	0.38	1.58	0.000	-1.09	0.000				
KU 5.1	17.15	70.72	0.40	1.55	0.000	-1.18	0.000				
KU 5.2	17.65	75.60	0.36	1.67	0.000	-0.96	0.000				
KU 5.3	18.15	78.26	0.17	1.59	0.000	-1.13	0.000				
KU 5.4	18.65	81.34	0.30	1.55	0.000	-1.19	0.000				
KU 5.5	19.15	80.32	0.01	1.54	0.000	-1.04	0.000				
KU 5.6	19.65	79.95	0.35	1.60	0.009	-1.00	0.067	-26.00	0.388	3.45	1.520
KU 5.7	20.15	79.89	0.34	1.54	0.000	-1.09	0.000				
KU 5.8	20.65	79.81	0.28	1.48	0.000	-1.19	0.000				
KU 5.9	21.15	81.07	0.29	1.50	0.000	-1.12	0.000				
KU 5.10	21.65	81.42	0.28	1.53	0.000	-1.05	0.000	-26.73	0.941	1.20	1.434
KU 5.11	22.15	81.52	0.26	1.51	0.000	-1.20	0.000				
KU 5.12	22.65	0.00	0.23	1.50	0.000	-1.17	0.000				
KU 5.13	23.15	79.51	0.32	1.51	0.000	-1.15	0.000				
KU 5.14	23.65	79.20	0.32	1.53	0.000	-1.12	0.000				
KU 5.15	24.15	76.31	0.35	1.52	0.000	-1.07	0.000				
KU 5.16	24.65	77.18	0.33	1.51	0.000	-1.02	0.000				
KU 6.1	25.15	77.71	0.01	1.64	0.000	-0.90	0.000				
KU 6.2	25.65	75.90	0.26	1.57	0.000	-0.96	0.000				
KU 6.3	26.15	73.92	0.45	1.56	0.070	-1.01	0.182	-24.43	0.029	8.84	0.736
KU 6.4	26.65	77.27	0.21	1.57	0.000	-1.00	0.000				
KU 6.5	27.15	79.53	0.01	1.43	0.000	-1.06	0.000				
KU 6.6	27.65	79.72	0.20	1.48	0.000	-1.09	0.000				
KU 6.7	28.15	78.73	0.01	1.52	0.000	-0.99	0.000				
KU 6.8	28.65	79.23	0.01	1.55	0.062	-0.90	0.028	-24.67	0.189	13.37	1.321
KU 6.9	29.15	76.51	0.42	1.52	0.000	-0.95	0.000				
KU 6.10	29.65	80.08	0.19	1.48	0.000	-1.00	0.000				
KU 6.11	30.15	83.02	0.01	1.48	0.000	-1.09	0.000				
KU 6.12	30.65	84.97	0.12	1.48	0.000	-1.19	0.000				
KU 6.13	31.15	84.17	0.14	1.40	0.000	-1.11	0.000				
KU 6.14	31.65	84.72	0.23	1.54	0.000	-1.16	0.000				
KU 7.1	32.15	85.15	0.18	1.40	0.000	-1.56	0.000				
KU 7.2	32.65	90.07	0.11	1.42	0.000	-1.88	0.000				
KU 7.3	33.15	89.06	0.21	1.52	0.047	-1.77	0.034				
KU 7.4	33.65	91.94	0.06	1.50	0.000	-1.67	0.000				
KU 7.5	34.15	93.69	0.10	1.48	0.000	-1.91	0.000				
KU 7.6	34.65	94.82	0.06	1.45	0.000	-2.15	0.000				
KU 7.7	35.15	94.03	0.19	1.50	0.000	-1.95	0.000				
KU 7.8	35.65	95.12	0.01	1.51	0.000	-1.96	0.000				
KU 7.9	36.15	97.19	0.01	1.50	0.000	-2.02	0.000				
KU 7.10	36.65	96.73	0.20	1.43	0.000	-2.13	0.000				
KU 8	40	96.36	0.11	1.52	0.014	-2.01	0.041	-25.10	0.300		
KU 9	45	98.85	0.01	1.71	0.026	-0.83	0.070				
KU 10	50	97.72	0.04	1.52	0.039	-1.58	0.043				

Table S7. Kultirenden, Denmark. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰), stable carbon isotope composition of bulk organic carbon ($\delta^{13}C_{org}$; ‰), and stable nitrogen isotope composition of bulk organic matter ($\delta^{15}N$; ‰).

Sample	cm	%$CaCO_3$	%TOC	$\delta^{13}C_{carb}$	st. dev.	$\delta^{18}O_{carb}$	st. dev.
HS -1	-5.00	97.18	0.11	1.82	0.03	-1.40	0.04
HS 0	0.00	41.47	0.17	1.75	0.00	-0.92	0.00
HS 1	0.50	11.86	0.12	1.26	0.03	-1.55	0.14
HS 2	1.50	36.04	0.07	1.61	0.02	-0.91	0.09
HS 3	2.50	60.22	0.03	1.69	0.09	-0.58	0.20
HS 4	3.50	68.42	0.21	1.54	0.00	-0.75	0.00
HS 5	4.50	77.23	0.20	1.62	0.03	-0.61	0.09
HS 6	5.50	82.88	0.16	1.71	0.04	-0.46	0.18
HS 7	6.50	95.75	0.11	1.69	0.03	-0.83	0.04

Table S8. Højerup, Denmark. Percentage of calcium carbonate (%$CaCO_3$), percentage of total organic carbon (%TOC), stable carbon isotope composition of bulk carbonate ($\delta^{13}C_{carb}$; ‰), and stable oxygen isotope composition of bulk carbonate ($\delta^{18}O_{carb}$; ‰).